B4Slider — User Manual

B4Slider — four-button camera slider panel

How to operate a **ready-configured** B4Slider on set.
App: `B4Slider.py` · config: `B4SliderConfig.py` (`B4S_*`).
Shared motion / LED stack: ../../api/overview.md. Installer hub: ../jkslider/technical/README.md.
One-page set card: cheat-sheet/cheat-sheet.pdf (HTML source).

B4Slider is a **minimal** UIC: **MOVE_L**, **MOVE_R**, **OPTION**, **SET**, one **SPEED** pot, and an RGB status LED. There is no keypad A/B/C, STOP key, DELAY, or TIMELAPSE. Soft travel limits **are** the A/B working window (see Workflow: A / B).

Optional second pot (**ACCEL**) when `B4S_USE_ACCEL_POT=1`. OLED is not required.

The panel Pico talks to a **motion board** (SliderMC) over UART, or to an MKS SERVO via MC_MKS_Client. If that link is unplugged, the UI may still start, but moves will not work — see Technical Manual — Link.

The stack **supports** an optional **2nd STEP/DIR axis** (`axis2_use`) — typical **linear travel + pan**, time-synced. **This panel** is a **1-axis** operator UI (`set_status_callback`). A custom 2-axis face uses `MC_Client` — UIC API. Wire: protocol.md.

Getting started

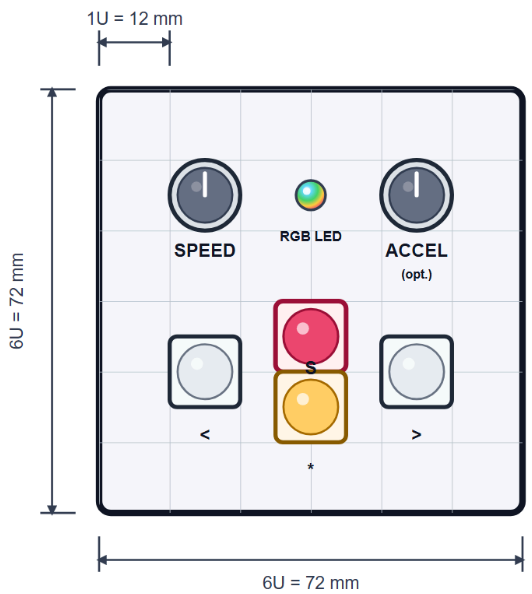
1. Power on — status LED does a rainbow while locked (if unlock is enabled).
2. **Unlock** — press **OPTION** (`*`). (Disable with `B4S_BOOT_UNLOCK = False`.)
3. Soft limits start at **full slider travel** (MC `slider_min` / `slider_max`). Nothing is remembered after power-off (no position file).
4. Dial **SPEED**, then use MOVE / SET as below.

OPTION is a modifier: hold it with another control. Alone it does nothing (except unlock at boot).

The **Key** column uses silk labels: `<` MOVE_L, `>` MOVE_R, `*` OPTION, `S` SET.

Panel layout

Recommended 6U × 6U (72 × 72 mm) discrete plate on a 12 mm grid. ACCEL is optional (`B4S_USE_ACCEL_POT`).



Silk: `S` SET, `<` MOVE_L, `>` MOVE_R, `*` OPTION.

Knobs

SPEED

- Left → slower floor; right → faster (finer at the low end; `B4S_SPEED_CURVE_GAMMA`).
- Can be changed while moving (unless OPTION is held for max-speed boost).
- Full-scale ceiling is the panel/MC max (`B4S_SPEED_MAX_MM_S` / MC `max_speed`).

ACCEL (optional)

Only if `B4S_USE_ACCEL_POT = 1`:

- Left → gentler; right → snappier (same idea as JKSlider ACCEL).
- When the ACCEL pot is enabled, **SET hold** gestures for accel presets / learn are **off**.

Without an ACCEL pot, use **SET** holds for presets **L** (low) / **H** (high) — not related to A/B soft ends.

Move

Cruise, jog, boost, stop, halt.

ACTION	KEY	RESULT
(MOVE_L / MOVE_R) tap ≤⅓ s	`<` / `>` tap	**Locked cruise** toward that soft end at SPEED until stop, reverse, SET, halt, or arrival
(MOVE_L / MOVE_R) hold >⅓ s	`<` / `>` hold	Hold-to-run: moves while pressed; soft-stops on release
Same **MOVE** tip while locked	`<` or `>` tip	Soft-stop
Opposite **MOVE** while cruising	`<` or `>`	Reverse / retarget to the other soft end
OPTION + (MOVE_L / MOVE_R) tap / hold	`*` + `<` / `>`	Same as MOVE, but at panel **max speed**
OPTION hold while moving	`*` hold	Boost to max speed; release → back to SPEED pot
SET while moving	`S`	Soft-stop
MOVE_L + MOVE_R	`<` `>`	**Halt** (emergency stop); driver disabled until any key
MOVE_L + MOVE_R hold ≥ 1 s	`<` `>` hold ≥ 1 s	Swap left/right
All four (L+R+OPTION+SET)	`<` `>` `*` `S`	Halt + **reset like power-up** (full soft limits, loop off, accel preset L)

Tap vs hold uses `B4S_MOVE_TAP_MS` (default **333 ms**). Left is toward decreasing position when `B4S_LEFT_IS_NEGATIVE` is True (default).

Soft limits (A / B window)

There are no separate PosA / PosB buttons. The two soft ends **are** the shot window:

ACTION	KEY	RESULT
SET + MOVE_L tap	`S` `<` tap	Set **soft_limit_L** = current position (white blip)
SET + MOVE_R tap	`S` `>` tap	Set **soft_limit_R** = current position (white blip)
SET + MOVE_L hold ≥ 1 s	`S` `<` hold ≥ 1 s	Reset soft_limit_L → full slider min
SET + MOVE_R hold ≥ 1 s	`S` `>` hold ≥ 1 s	Reset soft_limit_R → full slider max
SET + L + R hold ≥ 1 s	`S` `<` `>` hold ≥ 1 s	Reset **both** ends to full slider

Travel is allowed **only** between soft_limit_L and soft_limit_R. MOVE cruise targets those ends (the old "goto A / B").

SET and OPTION

ACTION	KEY	RESULT
SET tap (idle)	`S` tap	Nothing (avoids accidental presses)
SET hold ≥ 1 s (idle)	`S` hold ≥ 1 s	Accel **preset L** (low) — only if no ACCEL pot
SET hold ≥ 3 s (idle)	`S` hold ≥ 3 s	Accel **preset H** (high) — only if no ACCEL pot
SET hold ≥ 5 s (idle)	`S` hold ≥ 5 s	**Accel learn**: SPEED pot maps to accel while held; release latches into the active preset — only if no ACCEL pot
OPTION + SET tap	`*` `S` tap	Disable motor driver (dim orange LED)
OPTION + SET hold ≥ 1 s (idle)	`*` `S` hold ≥ 1 s	Toggle **loop** armed (single ↔ ping-pong between soft ends)
Any button while disabled	—	Re-enable driver

While holding SET for accel, the LED flashes **white once per second** so you can count 1 / 3 / 5 without a display. Preset L / H / learn confirm with violet flashes (1 / 2 / 3).

Loop: arming does **not** start motion. Next MOVE cruise starts; on arrival at a soft end the carriage auto-retargets to the other end until you stop (SET / same MOVE tip / halt / all-four).

Color codes

RGB status LED (shared `UIC_Base`). Docs use **percent**; API uses 0…255 — see API — RGB status LED.

COLOR / PATTERN	MEANING
Rainbow	Boot unlock (until OPTION)
Dim white	Idle, enabled, single-run
Dim white ↔ dim blue	**Loop armed** (idle)
Yellow	Accelerating or decelerating
Green	Moving at cruise speed
Green / yellow + ~30% blue	Near a soft end (`B4S_NEAR_SOFT_MM`, default 3 mm)
Green / yellow + 100% blue	At a soft end
+ ~10% blue while looping	Ping-pong running
Dim orange	Driver **disabled**
Red fast blink	Hard limit
Red blink	Homing (if used on the motion board)
Solid red	DRV_ERROR / EMO
White blip	Soft limit set or reset confirm
White flash /s	SET hold second counter
Violet ×1 / ×2 / ×3	Accel preset L / H / learn latched
Red flash (+ white ×3)	Halt / all-four power-up reset

Cheat card

One-page set card: cheat-sheet/cheat-sheet.pdf (HTML).

Workflow: normal moving

1. **Unlock** with OPTION.
2. Dial **SPEED** (and ACCEL pot or SET presets L/H if no ACCEL pot).
3. **Tap MOVE_L** / **MOVE_R** for a short burst move: release within ~⅓ s and the slider continues cruising toward that soft end until you stop it, reverse it, or reach the limit.
4. **Hold MOVE_L** / **MOVE_R** longer than ~⅓ s for hold-to-run. The carriage moves while the button is down; release stops it.
5. **Same-side tap while cruising** stops the cruise; **opposite-side tap** reverses direction. This makes the move buttons behave like a left/right lock-and-stop control rather than a raw jog.
6. **OPTION before MOVE_L / MOVE_R** changes the start condition: when OPTION is already down before the move starts, the slider launches with **max speed + max accel**.
7. **OPTION during movement** is a speed boost only: **speed goes to max_speed**, while **accel stays at the current pot/preset value** until OPTION is released.
8. If you want to keep the current pot accel but make the move faster, hold OPTION after the move is already running. If you want the full startup acceleration burst, press OPTION before the move.
9. **SET** while moving soft-stops the carriage. **MOVE_L + MOVE_R** is the emergency halt; any button re-enables the driver after a disabled state.
10. **All four** (MOVE_L + MOVE_R + OPTION + SET) resets the session like power-up and clears the soft-limit / loop state.

Workflow: A / B (soft limits)

Use the soft ends as your two shot marks.

1. **Unlock** — press OPTION if the LED is still rainbow.
2. **Open the window** (optional) — if limits were shrunk earlier: **SET + L + R** hold ≥ 1 s to restore full travel.
3. **Frame end A** — MOVE or hold-jog to the first pose. Press **SET + MOVE_L** tap → soft_limit_L saved (white blip).
4. **Frame end B** — move to the second pose. **SET + MOVE_R** tap → soft_limit_R saved.
5. **Rehearse** — dial SPEED; **MOVE_L** or **MOVE_R** tap to cruise to that end. Same tip stops; opposite tip reverses.
6. **Loop (optional)** — idle: **OPTION + SET** hold ≥ 1 s (LED white↔blue). Then MOVE tap; carriage ping-pongs until SET / same tip / L+R.
7. **Boost** — hold OPTION while moving, or start with OPTION+MOVE, for max speed.
8. **Reset ends** — SET+MOVE hold ≥ 1 s resets that side to full slider; or all-four for a full session reset.

Config entries

Edit `B4SliderConfig.py` defaults, or overlay via `SliderPins.py`:

```
B4Slider = { ... }      # consumed by B4SliderConfig.py
UIC_config = { ... }    # shared LED / OLED / camera (UIC_Base)
MC_config = { ... }     # UART / floors (MC_Client)
```

Panel (`B4S_*` / pins)

KEY	DEFAULT	MEANING
`PIN_BTN_MOVE_L` / `MOVE_R`	6 / 7	`<` / `>`
`PIN_BTN_OPTION`	13	`*`
`PIN_BTN_SET`	5	`S` (was STOP on JKSlider discrete map)
`PIN_POT_SPEED`	26	SPEED ADC
`PIN_POT_ACCEL`	27	ACCEL ADC if used
`B4S_USE_ACCEL_POT`	`0`	`1` = ACCEL pot on; disables SET accel holds
`B4S_ACCEL_PRESET_L` / `_H`	100 / 400	mm/s² presets (no ACCEL pot)
`B4S_SPEED_MIN_MM_S`	1.0	SPEED pot floor
`B4S_SPEED_MAX_MM_S`	100.0	Panel ceiling (also clamped by MC)
`B4S_ACCEL_MIN_MM_S2` / `_MAX_`	50 / 500	ACCEL pot range / learn range
`B4S_MOVE_TAP_MS`	333	Tap vs hold threshold (~⅓ s)
`B4S_LONG_PRESS_MS`	1000	≥ 1 s
`B4S_EXTRA_LONG_MS`	3000	≥ 3 s
`B4S_LEARN_HOLD_MS`	5000	≥ 5 s accel learn
`B4S_LEFT_IS_NEGATIVE`	`True`	Left toward decreasing mm
`B4S_NEAR_SOFT_MM`	3.0	Near-soft LED distance (also sets UIC warn)
`B4S_LOOP_BLUE_ADD`	26	~10% blue while looping (0…255)
`B4S_BOOT_UNLOCK`	`True`	Require OPTION before enable
`B4S_LED_FLASH_ON_MS` / `_OFF_`	80	Flash timing
`B4S_LED_BLIP_MS`	120	Soft-limit confirm blip
`B4S_LED_PINGPONG_MS`	600	Loop-armed white↔blue period

Shared LED (`UIC_config`)

KEY	DEFAULT	MEANING
`SOFT_LIMIT_WARN_MM`	10.0	Near-soft distance (B4Slider overwrites from `B4S_NEAR_SOFT_MM` at run)
`LED_SOFT_NEAR_BLUE_ADD`	76	~30% blue add (0…255)
`LED_SOFT_AT_BLUE_ADD`	255	100% blue add at soft end
`LED_DIM_WHITE` / `LED_DIM_ORANGE`	0.12	Idle / disabled duty (0…1)
`LED_BLINK_HARD_LIMIT_MS`	80	Hard-limit red blink half-period

Deep wiring, SliderMC, and library details: Technical Manual, API, ARCHITECTURE.